

Here is what Hack a Day had to say about VIPER on their Kickstarter page

"Part of getting IoT projects off the ground is overcoming the language barrier between humans who want to easily prototype complex ideas and hardware that wants specific instructions. A company called Things On Internet (TOI) has created VIPER, a system to easily program any Spark Core, UD00 or Arduino Due with Python."

"VIPER enables developing embedded and IoT solutions in a faster and highly re-usable way. Hence, we believe that VIPER will be the Android for the embedded world."

—Daniele Mazzei, **VIPER** co-founder and system integration designer



with specialised garbage collector to suit embedded operations.

VIPER converts local and global names to 16-bit integers instead of saving it as strings. While this saves resources spent by the MCU, names information is not available during runtime. Also, for time-constrained functions, you can write the fast code in C and call it from Python script.

Customisability. As the embedded machine being used is virtual, you can customise it to your application. Customisation referred to here is in terms of the number of drivers initialised, functioning of ports, level of optimisation and so on. This results in saving precious flash and random access memory.

VIPER tools. A compiler reads the input code and produces an optimised byte-code that can be understood by the VM. Optimisation here involves saving code objects on flash for direct access and analysing the code for static reachability. A VM builder allows you to configure and create a VM that suits your application better, for better functionality and to help you work within your constraints.

There is also a VIPER UP-linker that prepares the packages required by the VM for executing the expected design. All of these are integrated into the integrated development environment (IDE) to enable one-click execution of the entire process.

The latest release, VIPER 0.2.0.0009, also includes a photon Wi-Fi driver, Adafruit neopixel light emitting diode (LED) matrix driver, an infrared read-and-write control module, drivers for servomotor and thermal printer and an animation module.

Inclusion of these additional features promises an enhanced and smooth user experience.

What VIPER Suite is

A comprehensive development environment, VIPER Suite consists of three main components. With the help of these you can easily and quickly design your embedded system and integrate with sensors and the cloud. Let us see each of these components one by one.

VIPER IDE. A browser based dedicated development environment, the IDE provides a platform for developing your Python code and managing your boards. It includes a compiler, debugger and an editor, alongside tutorials and example projects for an easy learning experience. A code editor based on Code Mirror makes it easy to navigate through your code, while providing a comfortable user interface. The integrated debugger has built-in support for exceptions, which although not Python exceptions, are faster to raise and handle. The IDE also allows you to inspect low-level VIPER VM byte-codes.

A serial console allows you to control communication with ports on the board. Managing connected devices is easy, thanks to the board management toolbar. A registered user is also automatically connected to VIPER cloud, where she or he can

store data and access sessions from different devices/places, without affecting continuity. Thus, all you need to do is develop scripts locally using the IDE and sync it with the cloud.

VIPER VM. The aim of VIPER VM is to bring Python into the embedded world. It uses a sub-set of Python, neglecting features not required by the embedded world. What the VM lets you do is develop board-independent Python scripts that can be easily re-used and simulated. VIPER supports most high-level features of Python like modules, classes, multi-threading, callback, timers and exceptions. In addition to these, it allows one to use custom hardware-related features like interrupts, PWM and digital I/O.

VIPER app. This is how you access your object. The app provides an interface for all network- or Bluetooth-powered VIPER objects. Easily downloadable through Android, iTunes or Windows stores, the app allows you to control your object using your smartphone or tablet. On launching the app, it automatically displays a list of viperised devices in the vicinity, which you can then connect to.

Although VIPER is in Python, app interfaces are based on Hyper Text Mark-up Language (HTML) or Java, making it easy for you to connect to your MCU or mobiles. You do not need to write any separate code for Android or iOS.

How to start using VIPER

With a zero-install development environment, all you need to do is unpack the packages that come with VIPER. Or, you can simply load the necessities from this month's DVD and start using it right away.

Find example projects on VIPER hacker platform (www.hackster.io/viper). Clear your doubts that crop up by posting on their community forum (www.community.viperize.it). For anything else, get onto their website, www.viperize.it ●

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